

Polynomial Regression: Fitting Curves

Polynomial regression is a powerful technique that combines multiple linear regression with **feature engineering** to fit complex, **non-linear curves** to the data. It's used when a simple straight line is not a good fit for the underlying relationship between the features and the target.

Creating Polynomial Features

The core idea is to create new features by raising an original feature to different powers.

- **Example:** For a single feature x (like house size), you can create new features to build more complex models:
 - **Quadratic Model:** Use x and x^2 . The model becomes $f(x) = w_1x + w_2x^2 + b$. This can fit a parabolic curve.
 - **Cubic Model:** Use x , x^2 , and x^3 . The model becomes $f(x) = w_1x + w_2x^2 + w_3x^3 + b$. This can fit a more complex S-shaped curve.
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The Importance of Feature Scaling

When you use polynomial regression, **feature scaling becomes extremely important**.

- **Reason:** The ranges of the engineered features can become drastically different.
 - If x (size) ranges from 1 to 1,000...
 - Then x^2 (size squared) ranges from 1 to 1,000,000.
 - And x^3 (size cubed) ranges from 1 to 1,000,000,000.
 - **Impact:** Without scaling, these vastly different ranges will cause gradient descent to converge very slowly and inefficiently.
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Other Feature Choices

You are not limited to just polynomial terms. You can engineer other non-linear features based on your understanding of the data.

- **Example:** You could use the **square root** of a feature.
 - **Model:** $f(x) = w_1x + w_2\sqrt{x} + b$
 - The square root function increases but at a decreasing rate, which might be a suitable model for some datasets.

How to Choose Features: Deciding which features to create or use is a key part of building a good model. Later in the specialization, you will learn systematic ways to evaluate and select the best features for your problem.